

A Generative Field Framework for Measuring Spatial Priors in Image Models

A method for measuring spatial priors in generative models.

A kernel for measuring geometry in model-generated images

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A framework for detecting the spatial behaviors that current metrics cannot measure.

THE PROBLEM

Current metrics cannot detect spatial priors.

Current evaluation focuses on:

- text-image compliance
- object / attribute correctness
- feature-space realism
- semantic similarity

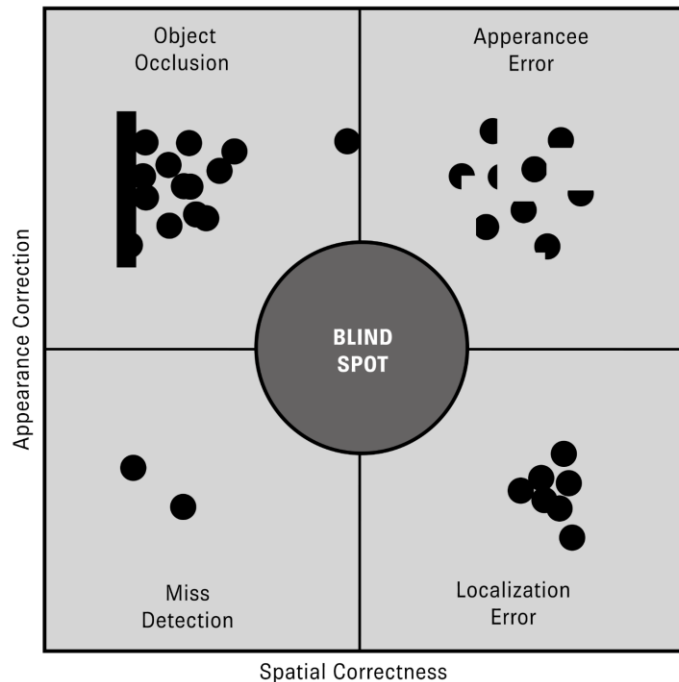
None of these metrics can answer:

- *How does the model structure?*
- *Where does structure begin to fail?*
- *What is the model's defaults?*
- *How does drift occur under perturbation?*

Spatial behavior is a blind spot.

This gap is not aesthetic. It is geometric.

This document introduces a minimal kernel for quantifying spatial priors across generative engines.



THE KERNEL (THE MINIMAL SET OF SPATIAL PRIMITIVES)

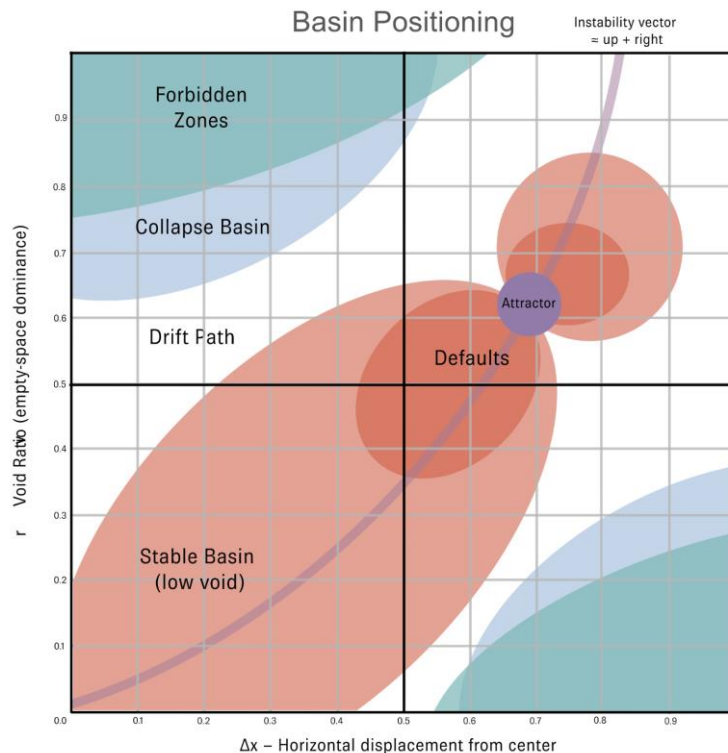
A spatial priors measurement using a small, model-agnostic kernel:

- Δx placement offset
- r_v void ratio
- ρ_r compression / packing density
- μ compositional cohesion
- x_p peripheral pull
- Optionally:
 - θ orientation stability
 - ds thickness/mass

These primitives project each frame into a **low-dimensional spatial field**, revealing:

- Attractors
- Compositional defaults
- Stable basin (low void)
- Drift paths
- Collapse basins
- Forbidden zones

This kernel is intentionally minimal: the smallest measurable set that explains the majority of spatial variance across engines.



CLAIM

Every generative engine has a geometry.

Models exhibit habitual spatial attractors: they do not merely place objects; they compose.

They operate inside a **spatial prior field**, a set of habitual attractors and avoided regions that govern:

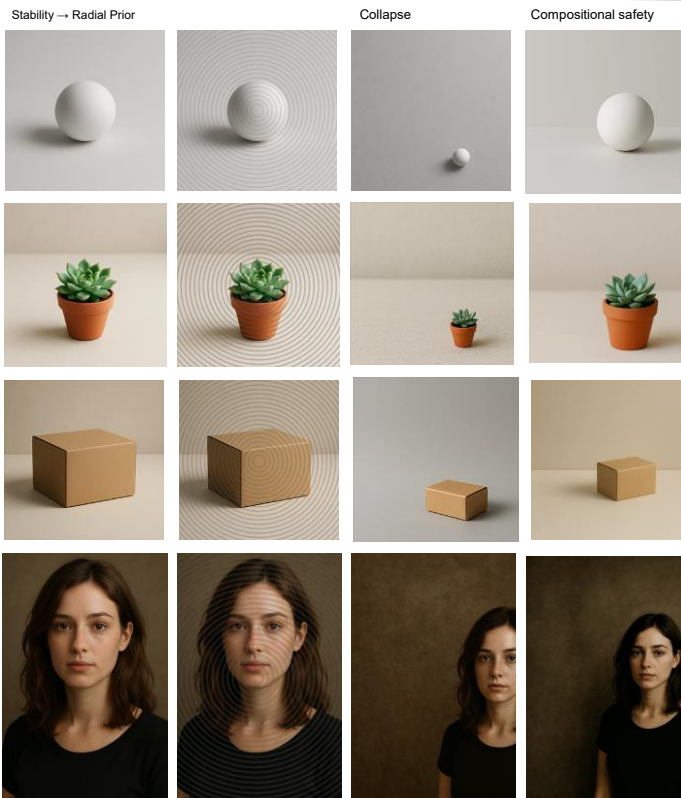
- **Stability** (ex. here radial prior)
- **Drift**
- **Geometric Collapse**
- **Compositional safety** (fallback)

The field is not inferred from training data; it is inferred from the model's generative behavior.

These behaviors come from the **model's spatial prior**, not its semantics.

The patterns shown here are measurable, repeatable, and consistent across runs.

Same object. Three spatial priors.



Stability reveals the attractor.

WHY SEMANTIC METRICS MISS THIS

Semantic success $\neq$ spatial success.

- Models can satisfy the prompt but still drift geometrically.
- Small spatial changes accumulate long before meaning breaks.

The kernel quantifies these shifts.

Structural collapse doesn't originate in meaning. It originates where the **spatial prior collapses**.

Structural instability is invisible to semantic metrics, but it is measurable in the spatial field:

- **Δx drift** (placement instability)
- **μ decay** (loss of cohesion)
- **p_r compensation** (mass redistribution)
- **early collapse onset** (geometry fails before content does)



"A photographic portrait of a young woman with natural lighting. Centered."
→ stable prior, balanced placement



"Regenerate smaller, off-center right."
→ placement obeys prompt, but geometry begins adjusting



"Make even smaller, near the right edge, maintain consistency."
→ model compensates: proportions shift, cohesion decays, void expands

EVIDENCE: SAME PROMPT → FOUR COMPOSITIONAL PRIORS

A single prompt → four spatial solutions.

This variation is geometric, not semantic. Each model expresses a distinct compositional prior.

- GPT + Sora share a right-biased minimal-void prior.
- MJ and OA diverge: compression vs shallow-tilt geometry.

Consequence:

Models don't just style differently, **they compose differently.**

The kernel makes those priors quantifiable.

Prompt: A single object placed near a wall, lit softly from one side. Neutral color palette. No decoration, no text, no additional objects. Photographic, simple, quiet.



Engine	Δx	Δy	r_v	ρ_r	μ	x_p	Quick Structural Read
GPT	0.69	0.43	0.61	0.70	0.70	0.28	Hard right-bias + high void. GPT pushes object low-right and keeps the field extremely empty.
Sora	0.61	0.55	0.77	0.58	0.58	0.38	Strong right-pull, high void. Matches GPT in Δx but even more empty. "Clinical minimalism."
MidJourney	0.30	0.47	0.44	0.69	0.69	0.26	Left-weighted, crowded, lowest void. MJ compresses the frame and prefers corner anchoring.
OpenArt	0.56	0.45	0.48	0.59	0.59	0.22	Right-leaning but denser than GPT. More conventional still-life packing, mild compression.

FIELD MAP (THE 4 BASINS)

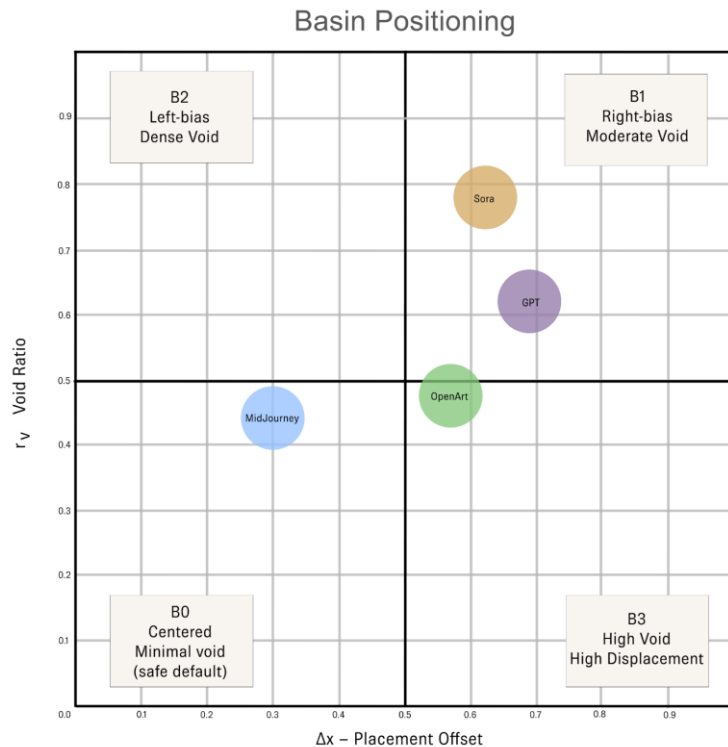
Each engine reveals a **different inductive bias**:

- placement bias (Δx)
- void usage (r_v)
- packing density (ρ_r)
- peripheral pull (x_p)
- compositional cohesion (μ)

These clusters remain stable across prompt families, confirming that the kernel is reading inductive bias rather than one-off stylistic artifacts.

This variation is not semantic.
It is geometric.

Each model composes differently.



Most models cluster in one or two basins regardless of prompt diversity, evidence of stable priors.

PERTURBATION DRIFT

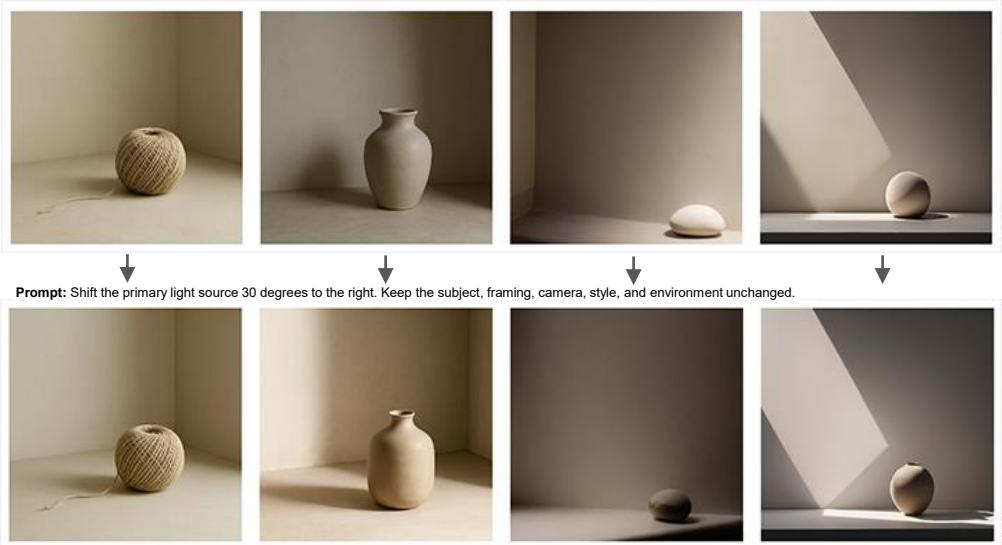
Under small prompt variation or intensity changes, models exhibit predictable drift:

- **GPT**: absorbs shift → shadow tightening, pocket retention
- **Sora**: centroid-locking, void tightening, downward energy pull
- **MidJourney**: global darkening, edge-packing, left-compression
- **OpenArt**: line hardening, tilt stabilization, pocket rigidity

Under small prompt variation or intensity changes, models exhibit predictable drift

Movement is not one metric direction.

Engines = **disproportionate structural correction**.



Metric	GPT	Sora	MidJourney	OpenArt
r_v Var (void-ratio variability)	Slight ↓	Moderate ↓	Large ↓	Moderate ↓
x_p -depth (peripheral pull)	↑ low	↓ inward	↓ mid-high	↓ low

PERTURBATION DRIFT (EARLIEST SIGNAL OF COLLAPSE)

Under small prompt variations, models show predictable drift.

- Void ratio (r_v) is often the earliest pre-collapse signal.
- Spatial structure degrades *before* semantic errors appear.
- Each architecture has a characteristic drift pattern, a geometric fingerprint.

Consequence

These patterns expose **measurable priors** in:

- object placement
- void allocation
- balance and mass distribution

They reveal how a model tries to **restore** its internal schema even as geometric collapse begins. This behavior invisible to semantic metrics.

These fingerprints reveal **measurable priors** in mass placement, void allocation, and balance.

Sora's drift is not linear → **geometric collapse** → **restoration** → **re-collapse**.



Step 1 (collapse onset):
 $\Delta X \downarrow, r_v \downarrow, \rho_r \downarrow, \mu \downarrow, x_p \downarrow$



Steps 2–3 (schema restoration):
 $\Delta X \uparrow, r_v \uparrow, \rho_r \uparrow, \mu \uparrow, x_p \uparrow$



Seeking its original right-bias prior

WHY CURRENT METRICS CAN'T SEE THIS

IS, FID, KID, CLIPScore, T2I-CompBench, GenEval all measure some combination of:

- semantic correctness
- object/attribute/relationship fidelity
- feature-space realism
- text satisfaction
- diversity

But none measure:

- spatial attractors
- boundary behavior
- drift under perturbation
- compression / cohesion
- void logic
- structural collapse

They evaluate *what* the model generated.

The kernel evaluates *how* the model composes.

This distinction is central: composition is a form of model reasoning, not merely a visual outcome.

Semantics can match while geometry diverges.

Semantically identical / Spatially divergent



Figure 1 - Woman Standing (Classic Centered)

$\Delta x = 0.022$ (nearly centered - 23px offset)
 $r_v = 0.962$ (high void - 86% empty space)
 $\rho_v = 0.330$ (low density - subject is compact)
 $\mu = 0.760$ (high cohesion)
 $x_p = 0.426$ (moderate-to-strong peripheral pull)

Figure 2 - Woman in Motion (Gestural Asymmetry)

$\Delta x = 0.003$ (centered - 3px offset)
 $r_v = 0.765$ (high void)
 $\rho_v = 0.261$ (low density - dispersed)
 $\mu = 0.763$ (high cohesion despite dispersion)
 $x_p = 0.402$ (good peripheral pull)

Figure 3 - Woman (Diagonal Tension)

$\Delta x = 0.048$ (slightly off-center - 49px offset)
 $r_v = 0.877$ (very high void - 88% empty)
 $\rho_v = 0.101$ (very low density - highly dispersed)
 $\mu = 0.774$ (high cohesion)
 $x_p = 0.469$ (strong peripheral pull - highest of the three)

APPLICATIONS

Different objects, same phenomenon: each model expresses a stable spatial bias even under a furniture prompt.



1. Version regression

Detect when a new release shifts composition or stability.

2. Evaluation

Add a missing dimension to generative benchmarks.

3. Safety

Identify structural failure earlier than semantic failure.

4. Model comparison

Fingerprint spatial priors across architectures.

5. Video

Measure temporal drift of spatial priors across frames.

6. Steering

Use kernel metrics to nudge models into or out of collapse basins.

Metric	GPT	Sora	MidJourney	OpenArt
Δx	0.18	0.34	0.11	0.39
Δy	0.30	0.27	0.19	0.36
r_v	0.52	0.64	0.41	0.71
ρ_r	0.60	0.49	0.72	0.46
μ	0.76	0.81	0.69	0.74
x_p -depth	0.24	0.33	0.21	0.37
Basin	B2	B3	B1	B3
Drift	up-right	diag up-right	down-right	corner-ward
Signature	stable, product-shot	cinematic edge-weighted	decorative / frame-filling	architectural, void-driven

SUMMARY

The kernel makes spatial priors measurable.

- Models compose with hidden geometry, and that geometry is measurable.
- Perturbation reveals drift and geometric collapse before semantics fail.
- This adds a missing dimension to evaluation and safety.

**Spatial reasoning is not a subjective aesthetic issue.
It is a measurable part of model behavior.**

Spatial priors operate independently from semantic correctness; understanding them is essential for robust, interpretable generative systems.

→ This framework is compatible with existing benchmarks and extends them into the spatial domain.



ABOUT

I am Rus.

25 years in drawing + compositional systems.

Marketing executive + Former Managing Director at R/GA (Amex, LEGO, Unilever, Verizon).

Independent researcher mapping spatial priors in generative models.

Demonstrations available upon request

Sample notebooks: field mapping, kernel measurement, attractor analysis.

Model comparisons: GPT vs Sora vs MJ vs OA.

Appendix: Notes on Assumptions and Reproducibility

1. Structural Coordinates, Not Benchmarks

The values shown (Δx , r_v , p_r , μ , x_p) are **structural coordinates**, derived from a consistent spatial-reading protocol.

They capture *ordering*, *basin identity*, *drift direction*, and *relative magnitude*, the phenomena under study.

They are **not presented as statistical performance metrics**.

2. Reproducible in Principle

Each coordinate corresponds to a simple geometric primitive:

- Δx , Δy — centroid offsets
- r_v — void ratio (empty vs occupied area)
- p_r — packing density / continuity
- μ — cohesion proxy
- x_p — peripheral-pull depth

A deterministic pipeline (mask $\rightarrow$ geometry $\rightarrow$ coordinates) will be released as part of the instrumentation packet.

3. Hybrid Computer-Vision Option

A vision model can identify the material cluster (object + shadow + penumbra), after which the kernels are computed deterministically.

4. Normalization

Values are normalized for clarity and comparability; raw kernel extraction will be available in the accompanying notebook.

